Abstract class & abstract method scenario :

```
                                    public abstract class Vehicle
                                    {
                                    public abstract void run();
                                    }

class Car extends Vehicle           class Bike extends Vehicle          class Bus extends Vehicle
{                                   {                                   {
public void run()                   public void run()                   public void run()
{                                   {                                   {
}                                   }                                   }
}                                   }                                   }
```

Points :

1) In Java, Whenever **action is common but implementaticies are different then we should use abstract method.**

2) If a class contains **at least one** method as an abstract method then that class will become as an abstract class. (We should declare that class as an abstract class) .

3) We cannot create an object for abstract class.

4) Overriding all the abstract methods of super class **is compulsory** in the sub classes otherwise sub-class will also become as abstract class hence Object will not be created

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Abstract class and abstract methods :

A class that does not provide complete implementation (partial implementation) is defined as an abstract class

An abstract method is a commonsgenric method which is used to provide easiness to the programmer because instead of writing many methods we can write only one method.

An abstract method observation is very critical because every abstract method contains abstract keyword, abstract method does not contain any method body and at the end there must be a terminator ; (semicolon)

In Java, whenever action is common but implementations are different then we should use abstract method, Generally we declare abstract method in the super class and its implementation must be provided in the sub classes.

If a class contains at least one method as an abstract method then we should compulsory declare that class as an abstract class.

Once a class is declared as an abstract class we can't create an object for that abstract class.

Till the abstract methods declared in the super class must be overridden in the sub classes otherwise the sub class will become as an abstract class hence object can't be created for the sub class as well.

In an abstract class we can write all abstract methods OR all concrete methods or combination of both the methods.

It is used to achieve partial abstraction that means by using abstract classes we can achieve partial abstraction in Java.

The abstract class may or may not have abstract method but an abstract method must have abstract class.

Note :- We can't declare an abstract method as final, private and static (Illegal combination of modifiers)

We can't declare an abstract class as a final.

LPrograms:

```
sealed abstract class Shape permits Square, Rectangle, Circle
{
    public abstract void draw();
}
non sealed class Square extends Shape
{
    @Override
    public void draw();
    {
        SS.println("Drawing Square");
    }
}
final class Rectangle extends Shape
{
    @Override
    public void draw()
    {
        SS.println("Drawing Rectangle");
    }
}
final class Circle extends Shape
{
    @Override
    public void draw();
    {
        SS.println("Drawing Circle");
    }
}
public class AbstractDemo1
{
    public static void main(String[] args)
    {
        Shape shape = null;
        shape = new Square(); shape.draw();
        shape = new Rectangle(); shape.draw();
        shape = new Circle(); shape.draw();
    }
}
```

Q:

```
abstract class Bike
{
    protected int speed = 90;

    public Bike()
    {
        SS.println("Bike class constructor");
    }

    public void getBikeDetails()
    {
        SS.println("It has two wheels & one petrol engine");
    }

    public abstract void run();
}

class Honda extends Bike
{
    @Override
    public void run()
    {
        SS.println("Honda bike is running");
    }
}

public class AbstractDemo2
{
    public static void main(String[] args)
    {
        Bike obj = new Honda();
        obj.getBikeDetails();
        obj.getSpeed();//90
        obj.run();//Honda
    }
}
```

Note : abstract class constructor will be executed through sub class object by using super()

Q:

What is the advantage of writing instance variable/Object properties/ NSF(), constructor and non static block inside an abstract class, if we can't create an object for abstract class ?

Yes, We can't create an object for abstract class but still we can write instance variable, constructor and non static block inside abstract class.
abstract class also contains Object properties and behavior but these properties will initialize through SUB CLASS OBJECT using super() so we can write object properties, non static block as well as constructor inside an abstract class.